

Relating PVDF Piezo Voltage to PVDF Piezo Position on a Cantilever with Circular Cutouts

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Introduction

Oil and natural gas are primarily transported by pipelines [1]. Monitoring these pipelines is crucial to detect cracks and corrosion that could reduce transport efficiency and lead to devastating leaks. Sensors that detect pipeline defects must be placed inside the pipe, which makes them difficult to power. Light cannot enter the pipe, so solar panels cannot directly supply energy. Batteries must be replaced, which means extra maintenance procedures for pipelines. Thus, the best approach would be to convert the kinetic energy of flowing fluid into electrical energy. Piezoelectric materials placed on cantilevers are a promising method for this kinetic-to-electrical energy conversion [2].

A cantilever can easily oscillate in the presence of flowing fluid if two prongs are attached to the free end to interact with the fluid. A flexible piezoelectric material, such as polyvinylidene fluoride (PVDF), can bend with the oscillation of a cantilever. PVDF has dipoles on the microscopic scale, but they cancel out due to symmetry when the PVDF is in its relaxed state. However, deformation changes the distance and angles between dipoles, causing a net polarization. This polarization induces a voltage that can generate electrical energy [4]. The oscillatory motion of a cantilever produces an AC voltage, and a simple rectifier circuit can convert the AC voltage into a DC voltage to power devices [5].

The piezoelectric material on the cantilever acts as a mass-spring-damper system, commonly referred to as a single-degree-of-freedom energy harvester (SDOF). The mass at the end of the cantilever moves in only one direction, hence the term "single degree of freedom." The cantilever itself acts as the spring due to its ability to store potential energy when subjected to an external force. SDOFs only generate an appreciable amount of energy when the ambient frequency matches the system's resonance frequency [6]. The environment inside a pipeline can vary, making it challenging to continuously achieve a narrow resonance frequency. Fortunately, the piezo on a cantilever is a type of broadband system, meaning it generates a reasonable amount of energy over a wide range of frequencies.

Although a cantilever is a simple object, many of its characteristics significantly affect its ability to bend the piezo and generate voltage. A cantilever's shape is of particular interest because it affects strain distribution, coupling efficiency, resonance frequency, and mode shapes. For example, rectangular cantilevers concentrate strain at the root, rendering the rest of the cantilever inefficient for piezoelectric energy production [2]. However, simulations and measurements have proven that strain can be distributed more evenly by tapering the cantilever. One promising cantilever shape features elliptically shaped cavities along its length to distribute strain in the thinner regions. These strain distribution properties may increase a cantilever's total energy-producing surface area for piezoelectric materials.

Cantilever Design

The goal was to investigate the relationship between a PVDF piezo's voltage and the position of the piezo on a cantilever with circular cavities. The cantilever used in the piezo position experiments had the characteristics depicted in Figure 1.

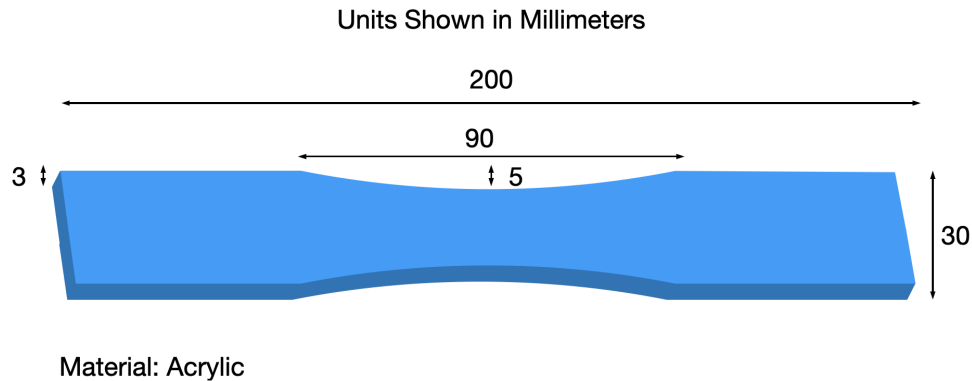


Figure 1: *Dimensions of cantilever used in the piezo placement experiments*

The cantilever was designed in Autocad. A 200x30 mm rectangle with dimensions corresponding to the cantilever length and width was created with the line tool. Then, three points were placed on the rectangle to define the cavity as shown in Figure 2. The two outer points defined the cavity length. The inner point defined the cavity depth. A circle that perfectly intersected these three points was made, and its image was cut away from the rectangle to produce the cavity. This method of creating the cavity was inspired by AutoCad 101 [8].

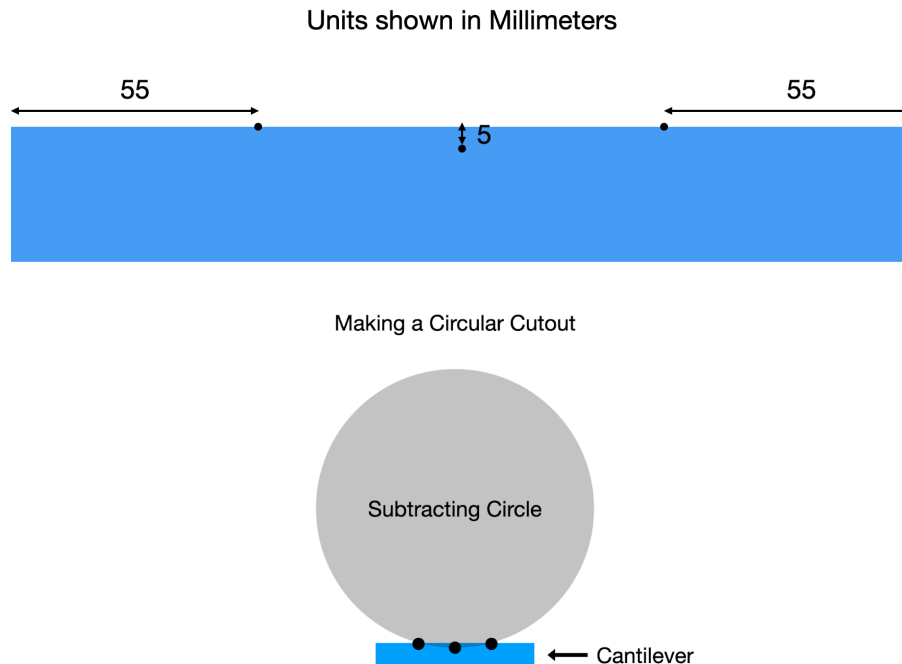


Figure 2: *Creating the cantilever cavity*

Additionally, one 15x10 mm rectangle and two 2 mm radius holes were cut away from the rectangle to accommodate the prongs. Similar cuts were made at the opposite end of the cantilever for the anchor, except the holes were 1.5 mm in radius, and the rectangle was 15x7 mm. The resulting cantilever design is shown in Figure 3.

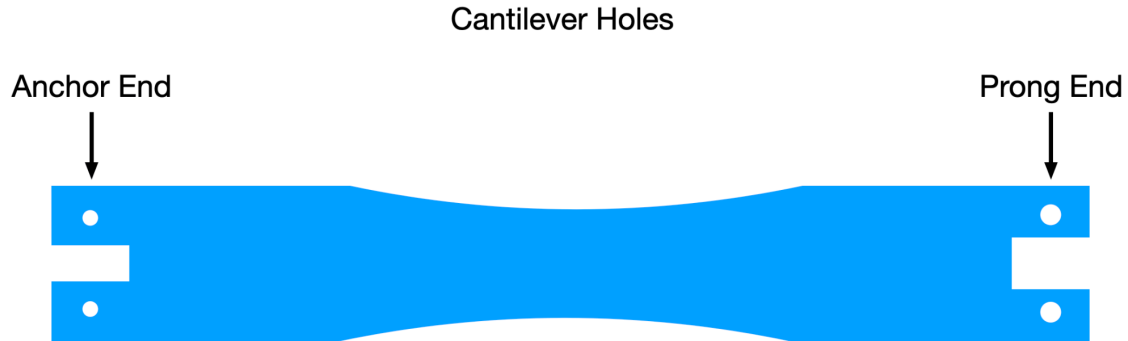


Figure 3: *Cantilever with holes and rectangular cuts to accommodate the prongs and anchor*

Prong Design

Prongs were designed on a blender. Two slabs of thickness 4 mm and distance 2 mm from each other were connected by a 15x10 mm rectangle at the middle so that they perfectly fit onto the prong end of the cantilever. These slabs had two prongs connected to them at the end. Each prong was 45 mm long and was 120 degrees apart from the other prong. The slabs also had 2 mm radius holes, just like the ones on the cantilever prong end.

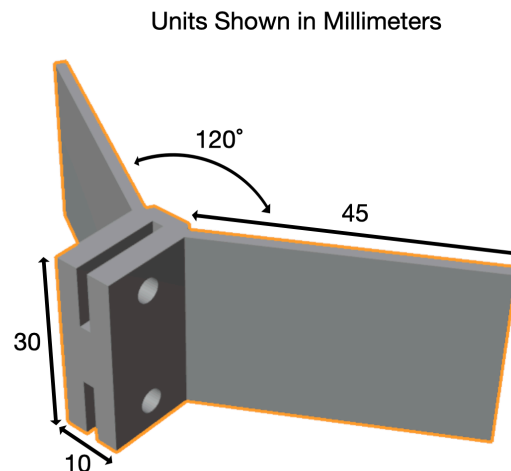


Figure 4: *Dimensions of cantilever used for piezo placement experiment*

Experimental Environment

The cantilever was secured to the tip of an optical post using two plexiglass clamps and bolts. The post was then tightened onto the center of an optical breadboard. This configuration was then enclosed in a rectangular, plexiglass wind tunnel. The wind tunnel was connected to a fan via a flexible aluminum duct pipe. This duct pipe was compressed to avoid generating turbulence within the pipe. Any exposed wires were covered in insulating tape to avoid shorting the circuit.

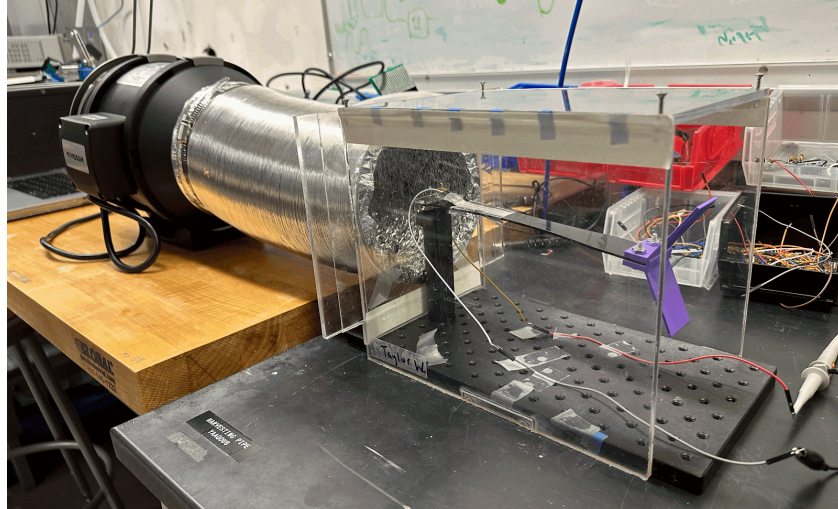


Figure 5: *Complete experimental setup*

Piezo Placement

The “cantilever point of measurement” is defined as the points along the anchor edge facing the cantilever. The “piezo point of measurement” is defined as the points along the piezo edge with the anode and cathode wires. The distance between both points was the distance marked for data. Once the correct distance along the cantilever was found, the piezo was secured to the cantilever using scotch tape.

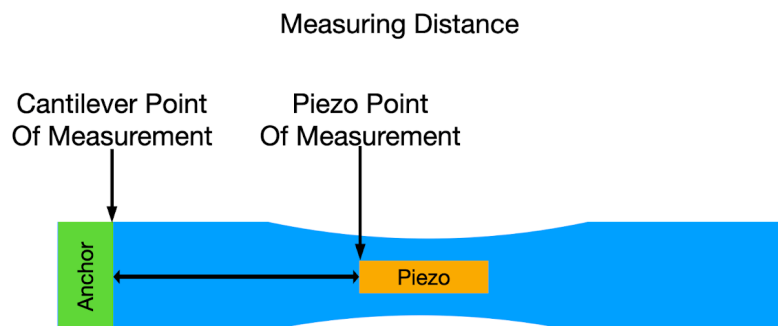


Figure 6: *Method for measuring distance marked for data collection*

Circuitry

The piezo generated very low voltages when oscillating on the cantilever. The voltage signal was amplified using an lm741 op-amp to better distinguish the effects of piezo location on piezo voltage. The feedback resistors amplified the piezo signal in a controlled manner, while the feedback capacitor smoothened the amplified signal. This circuit is shown in Figure 7 and was inspired by Tomasz Starecki [9]. The data was collected by probing between the op-amp output and ground using an oscilloscope.

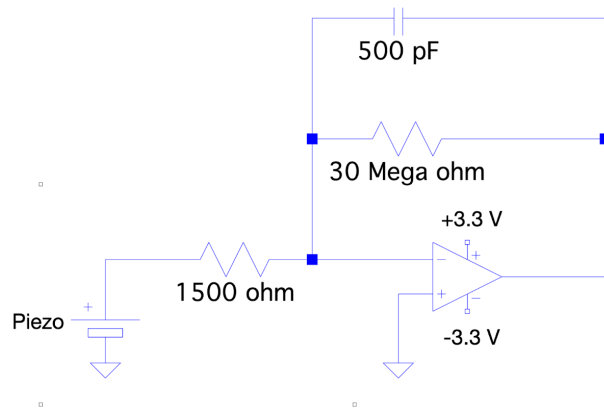


Figure 7: Circuit used to amplify piezo voltage signal. An lm741 op-amp from Texas Instruments was used.

Data

Piezo voltage signals were collected for piezos placed 1 to 10 cm away from the anchor edge. The free end of the cantilever was pushed down until the prong made contact with the optical breadboard, as shown in Figure 8. The cantilever was then quickly released to allow it to oscillate. Data was collected throughout the cantilever's oscillation. The voltage signal resembled an AC signal, and its amplitude steadily declined to 0 V as time passed. The piezo's maximum voltage amplitude steadily increased from 188 mV at 1 cm to 200 mV at 3 cm. The piezo placed at a distance of 4 cm exhibited the highest voltage output at 225 mV. The maximum voltage then steadily decreased from 203 mV at 5 cm to 184 mV at 7 cm. At 8 cm, the voltage sharply fell to 130 mV and continued to decrease rapidly until the 10 cm mark. The remaining figures represent the same data in different ways.

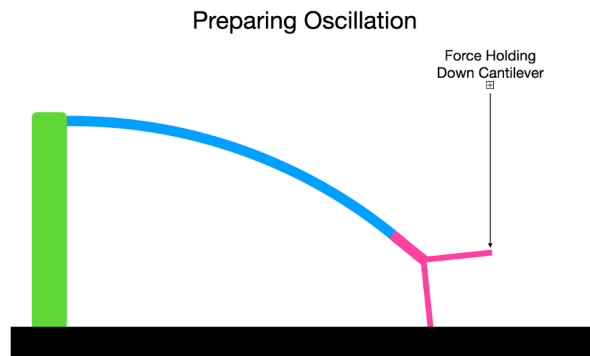


Figure 8: Configuration of bent cantilever before tension released for oscillation

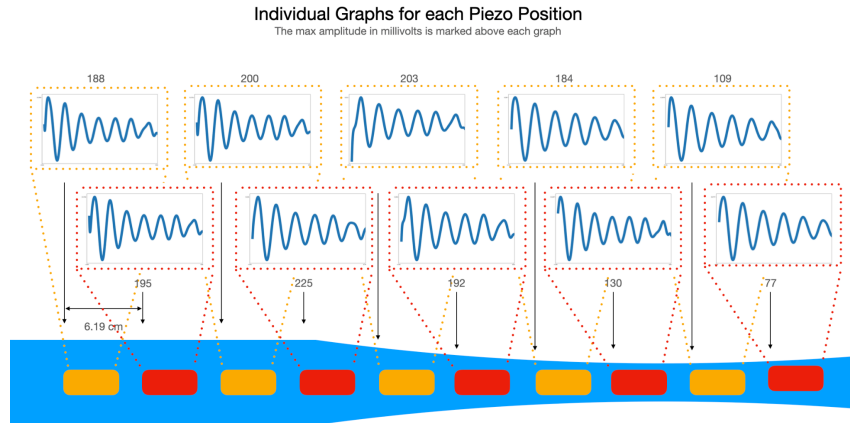


Figure 9: Oscilloscope data obtained at each piezo position. The dashed lines surrounding each image point to the position of the piezo corresponding to the data. Only the max amplitude is indicated for each piece of data.

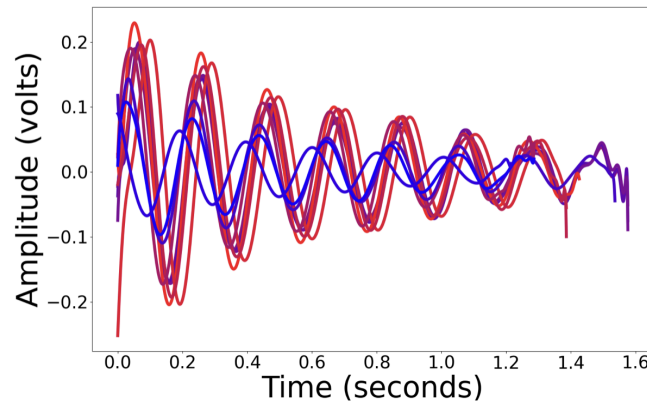


Figure 10: Combined oscilloscope data for each piezo position. The red oscillations indicate greater amplitude, while the blue oscillations indicate lesser amplitude.

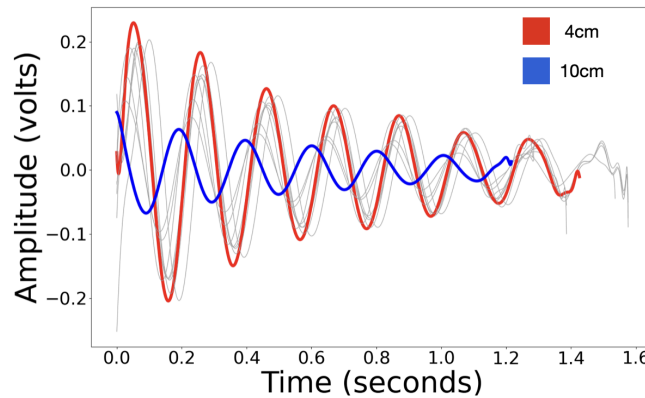


Figure 11: Combined oscilloscope data for each piezo position. Only the piezo positions with the greatest and least amplitude are colored for clarity.

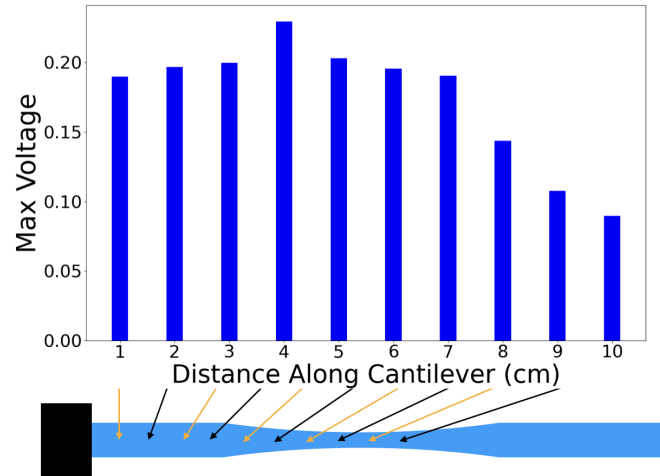


Figure 12: Bar graph indicating the max amplitude at each piezo position.

Discussion

The voltage oscillated around 0 V, and the frequency remained constant at 5 Hz, indicating that the PVDF's piezoelectric properties remained unchanged even after stress was applied. Additionally, the voltage oscillation closely resembled the cantilever's oscillation, demonstrating a high correlation between the voltage generated by the piezoelectric material and the cantilever's strain/bending.

Our data showed that a piezoelectric element placed 4 cm away from the anchor generated the most energy, with a peak output of 240 mV. Interestingly, 4 cm also marked the start of the cavity region. Since there was a high correlation between piezo voltage and cantilever strain, the start of the cavity region most likely experienced the highest strain. Rajinder Deol's Figure 5d supports this: He found that the strain along a cantilever slightly increased at the start of a cavity region. Figure 5d showed that the increase in voltage and strain was primarily due to additional strain generation at the cavity, rather than strain shifting from the anchor. However, Deol's data showed a decrease in strain from the anchor to the cavity region, seemingly contradicting our observed voltage increase from 185 mV at 1 cm to 213 mV at 3 cm [6]. It is important to note that the 6.7 cm long piezo extended from the marked distance toward the cantilever's free end. Thus, at the 1 cm mark, the piezo spanned 1 cm to 7.7 cm, overlapping the cavity region. Moving the piezo towards the free end meant it covered more of the higher-strain cavity region, explaining the voltage increase from 1 cm to 3 cm.

The strain in rectangular cantilevers rapidly decreases from the anchor to the free end. However, the cavity cantilever's slow voltage decrease from 5 to 7 cm suggested effective strain redistribution in thinning regions. Hao Wu confirmed this by measuring the strain of a triangular cantilever, which was capable of near-perfect strain redistribution. The anchor of the triangular cantilever experienced 3.75×10^{-5} units of strain, while the free end experienced 3.95×10^{-5} units. This is compared to a rectangular cantilever, where the strain was measured to be 3.50×10^{-5} units at the anchor and almost 0 units at the free end [10]. The triangular cantilever experienced significantly less change in strain along its length due to its ability to spread out the strain. Similarly, our cantilever's 5 to 7 cm region, which thins out like the triangular cantilever, exhibited strain redistribution properties. Conversely, the voltage sharply

declined after 8 cm, where the cantilever began to widen. Thus, while thinning widths facilitate strain redistribution, widening obstructs it.

Conclusion

Thinning cantilever portions generated more total strain and facilitated strain redistribution. As a result, the strain was likely highest at the thinning area of the cantilever with circular cavities. Piezoelectric materials placed in highly strained areas generated a higher voltage.

Extending the relatively long piezoelectric element from the anchor to the free end convoluted the relationship between voltage and strain. Therefore, it would be better to use a shorter piezoelectric material so the piezo distance from the anchor directly corresponds to the strain of a cantilever at a certain point. Cantilever strain should be measured alongside piezo output voltage to directly relate the two variables. Strain can be measured using a fiber Bragg grating, the same technology Hao Wu employed to measure the strain of a triangular cantilever [6].

Future studies should vary cavity length, cavity depth, cavity shape, prong shape, prong angle, and cantilever material to design the optimal cantilever for piezoelectric energy production.

Acknowledgements

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